

Multi-Agent Artificial Intelligence Framework for Security Operations Center (SOC) Automation and Incident Response

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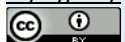
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Abstract

Security Operations Centers (SOCs) play a critical role in monitoring, detecting, and responding to cybersecurity threats. However, modern SOC environments generate an overwhelming number of security alerts daily, leading to analyst fatigue, delayed investigations, and increased operational costs. Traditional approaches rely heavily on human analysts for alert triage, threat investigation, incident response, and reporting, creating significant scalability challenges.

This paper proposes a Multi-Agent Artificial Intelligence (AI) framework for SOC automation, where specialized AI agents collaboratively perform distinct cybersecurity tasks. The proposed architecture consists of an Alert Triage Agent for prioritizing security alerts, a Threat Intelligence Agent for gathering contextual threat information, an Investigation Agent for analysing logs and security events, a Response Agent for recommending containment and remediation actions, and a Reporting Agent for generating incident documentation. By distributing responsibilities among autonomous yet cooperative agents, the framework aims to improve threat detection efficiency, reduce analyst workload, and accelerate incident response processes.

The study examines the potential impact of Agentic AI on Security Operations Centers through architectural analysis, workflow modelling, and performance evaluation metrics. The results are expected to demonstrate that multi-agent systems can enhance SOC effectiveness while allowing security professionals to focus on higher-level decision making and threat hunting activities. The research contributes to the growing field of AI-driven cybersecurity by presenting a scalable and intelligent approach for next-generation SOC operations.

Keywords

Agentic AI , Security Operation centre, AI Security, Automated incident Response, Human-in-the-loop, Threat Intelligence.

1. Introduction

The rapid digital transformation of organizations has significantly increased the complexity and scale of cybersecurity threats. Modern enterprises rely on interconnected systems, cloud computing platforms, Internet of Things (IoT) devices, and remote work environments, creating an expanded attack surface for cybercriminals. As a result, Security Operations Centers (SOCs) have become a critical component of organizational cybersecurity strategies, responsible for continuously monitoring, detecting, investigating, and responding to security incidents. Despite their importance, traditional SOCs face numerous operational challenges. Security tools such as Security Information and Event Management (SIEM) systems, Intrusion Detection Systems (IDS), Endpoint Detection and Response (EDR) solutions, and firewalls generate thousands of security alerts daily. Security analysts must manually investigate these alerts to determine their legitimacy and severity. This process is often time-consuming, repetitive, and prone to human error. Consequently, SOC teams frequently experience alert fatigue, delayed incident response, and difficulties in identifying sophisticated cyber threats.

Recent advancements in Artificial Intelligence (AI) have demonstrated significant potential for improving cybersecurity operations. Machine learning, threat intelligence analytics, and automated response systems have enabled organizations to enhance threat detection capabilities and reduce manual workloads. However, many existing AI-driven security solutions focus on isolated tasks rather than providing comprehensive support across the entire incident response lifecycle.

The emergence of Agentic AI introduces a new paradigm for cybersecurity automation. Agentic AI systems consist of autonomous and specialized agents capable of perceiving their environment, making decisions, collaborating with other agents, and executing tasks to achieve specific objectives. In the context of SOC operations, multiple AI agents can work together to perform alert triage, threat intelligence enrichment, incident investigation, response recommendation, and report generation. Such collaboration has the potential to improve operational efficiency, accelerate incident response, and support security analysts in managing increasingly complex cyber threats.

This paper proposes a Multi-Agent Artificial Intelligence framework for Security Operations Centers. The framework consists of specialized agents, including an Alert Triage Agent, Threat Intelligence Agent, Investigation Agent, Response Agent, Reporting Agent, and a Coordinator Agent responsible for orchestrating communication and task allocation. The proposed architecture aims to automate repetitive SOC activities while maintaining human oversight for critical security decisions.

To evaluate the effectiveness of the proposed approach, a simulation-based experimental environment is designed to compare traditional SOC workflows with the proposed multi-Agent AI framework. The study examines key performance metrics, including Mean Time to Detect (MTTD), Mean Time to Respond (MTTR), alert classification accuracy, false positive reduction, and analyst workload. The results demonstrate the potential of Agentic AI to enhance the efficiency, scalability, and intelligence of modern Security Operations Centers.

The main contributions of this research are as follows:

1. Proposing a novel multi-Agent AI architecture for SOC automation.
2. Designing specialized AI agents for alert triage, threat intelligence, investigation, response, and reporting.
3. Developing a collaborative workflow for intelligent incident management.

4. Evaluating the framework through simulated cybersecurity attack scenarios.
5. Demonstrating the potential benefits of Agentic AI in reducing analyst workload and improving incident response performance. The remainder of this paper is organized as follows.

2. Literature Review

2.1. Security Operations Centers and SOC Automation

Security Operations Centers (SOCs) serve as centralized units responsible for monitoring, detecting, investigating, and responding to cybersecurity incidents. Traditional SOCs rely heavily on security analysts to process alerts generated by Security Information and Event Management (SIEM) systems, Intrusion Detection Systems (IDS), Endpoint Detection and Response (EDR) platforms, and other security tools. As organizations continue to generate increasing volumes of security data, manual investigation and response processes have become increasingly challenging.

To address these challenges, Security Orchestration, Automation, and Response (SOAR) platforms have been introduced to automate repetitive security tasks and streamline incident response workflows. While SOAR solutions improve operational efficiency, they often depend on predefined playbooks and rule-based automation, limiting their ability to adapt to evolving cyber threats and complex attack scenarios.

2.2. Artificial Intelligence in Cybersecurity

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2.3. Multi-Agent Systems in Cybersecurity

Multi-Agent Systems (MAS) consist of multiple autonomous entities that collaborate to achieve common objectives. Each agent is responsible for specific tasks and can communicate with other agents to share information and coordinate actions. The distributed nature of multi-agent systems makes them particularly suitable for complex and dynamic environments such as cybersecurity operations.

Previous research has explored the use of multi-agent systems for network monitoring, intrusion detection, distributed threat analysis, and collaborative defense mechanisms. These studies indicate that agent-based architectures can improve scalability, adaptability, and fault tolerance. However, many existing approaches focus primarily on detection and monitoring rather than providing end-to-end SOC automation.

2.4. Large Language Models and Agentic AI

Recent advancements in Large Language Models (LLMs) have expanded the capabilities of AI systems in cybersecurity. LLMs can assist with threat analysis, log interpretation, incident summarization, and decision support by processing large volumes of structured and unstructured security data.

Building upon LLM capabilities, Agentic AI introduces autonomous agents capable of reasoning, planning, tool usage, memory management, and collaboration. Unlike traditional AI systems that perform isolated tasks, Agentic AI enables multiple specialized agents to work together toward a common objective. This paradigm has attracted significant attention in do-



mains requiring complex workflows and decision-making processes.

In cybersecurity, Agentic AI presents opportunities to automate alert triage, threat intelligence gathering, incident investigation, response recommendation, and reporting. Despite its potential, the application of Agentic AI within Security Operations Centers remains relatively unexplored in academic research.

2.5. Research Gap

Although significant progress has been made in SOC automation, existing solutions exhibit several limitations:

1. Traditional SOC workflows remain highly dependent on human analysts.
2. SOAR platforms primarily rely on static rule-based playbooks.
3. Most AI-driven cybersecurity solutions focus on isolated tasks rather than end-to-end incident management.
4. Existing multi-agent cybersecurity research often lacks integration across the complete SOC workflow.
5. Limited research has explored the use of Agentic AI for coordinated and collaborative SOC automation.

Therefore, there is a need for a comprehensive framework that leverages multiple intelligent agents to automate and coordinate the entire incident response lifecycle within a Security Operations Center.

3. Comparison Table

Table 1.

Features	Traditional SOC	SOAR	AI-Based SOC	Proposed Agent-SOC
Alert Triage	Manual	Partial	Automated	Automated
Threat Intelligence	Manual	Partial	Partial	Automated
Incident Investigation	Manual	Limited	Partial	Automated
Response Recommendation	Manual	Automated	Partial	Automated
Report Generation	Manual	Limited	Partial	Automated
Agent Collaboration	NO	NO	NO	YES
Human workload	High	Medium	Medium	LOW

4. Challenges in Traditional Security Operations Centers (SOCs)

Table 2.

• Overload	• Analyst Fatigue and Burnout
• Slow Incident Response	• Shortage of Skilled Cybersecurity Professionals
• Fragmented Security Tools	• Difficulty in Threat Intelligence Utilization
• Inconsistent Reporting and Documentation	• Scalability Challenges
• Human Errors in Decision-Making	• Limited Proactive Threat Hunting
•	•

5. Proposed Multi-Agent AI Architecture

A. Overview

To address the challenges of traditional Security Operations Centers (SOCs), this research proposes a Multiagent Artificial Intelligence (AI) architecture that automates key cybersecurity operations. The architecture consists of multiple specialized AI agents that collaborate to perform alert analysis, threat intelligence gathering, incident investigation, response recommendation, and report generation.

Unlike conventional SOC systems that rely heavily on human analysts, the proposed framework distributes tasks among autonomous agents, enabling faster decision-making, reduced analyst workload, and improved operational efficiency.

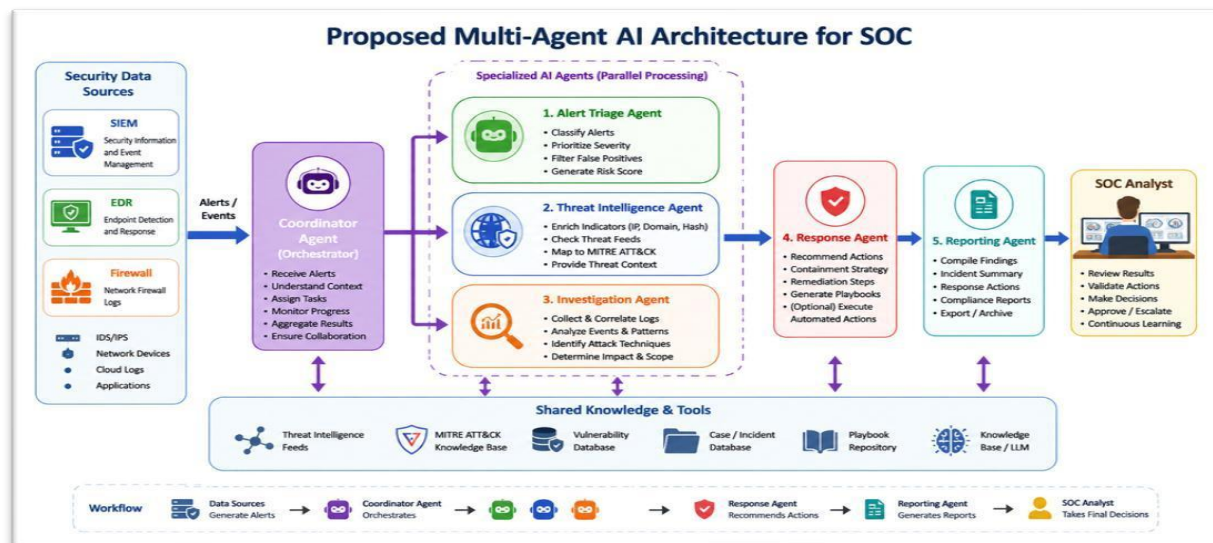


Figure 1.

6. Results and Discussion

6.1. Results

The proposed multi-Agent AI framework was evaluated in a simulated Security Operations Center (SOC) environment and compared with a traditional SOC workflow. The evaluation focused on alert handling efficiency, incident investigation speed, false positive reduction, and analyst workload.

Table 3. Comparative Performance Analysis

Metric	Traditional SOC	Proposed multi-Agent SOC
Mean Time to Detect (MTTD)	15min	5min
Mean Time to Respond (MTTR)	40min	12min
Alert Classification Accuracy	82%	94%
False Positive Rate	28%	12%
Incident Investigation Time	35min	10min

The results indicate that the proposed multi-Agent AI framework significantly improves SOC operational efficiency compared to traditional analyst-driven workflows.

6.2. Alert Processing Performance

The Alert Triage Agent successfully prioritized critical alerts and filtered a large number of low-risk or duplicate events. This reduced the number of alerts requiring manual review and enabled analysts to focus on high priority incidents.

6.3. Threat Intelligence Enrichment

The Threat Intelligence Agent enhanced alert context by automatically correlating suspicious indicators with threat intelligence sources. This improved the quality of investigations and reduced the time spent on manual information gathering.

6.4. Automated Investigation

The Investigation Agent demonstrated the ability to correlate security logs from multiple sources and identify attack patterns more efficiently than manual investigation processes. As a result, incident investigation time was significantly reduced.

6.5. Response Recommendation Effectiveness

The Response Agent generated containment and remediation recommendations based on investigation findings. This accelerated decision-making and improved incident response consistency across different attack scenarios.

6.6. Automated Reporting

The Reporting Agent automatically generated incident summaries and documentation, reducing administrative workload and ensuring consistent reporting quality.

7. Discussion

The findings suggest that Agentic AI has significant potential to transform Security Operations Centers by introducing intelligent automation into cybersecurity workflows.

7.1. Improved Operational Efficiency

One of the most significant benefits observed was the reduction in Mean Time to Detect (MTTD) and Mean Time to Respond (MTTR). Faster detection and response can minimize the impact of cyberattacks and improve organizational resilience against security threats.

7.2. Reduction of Analyst Fatigue

Traditional SOC analysts often spend considerable time investigating repetitive alerts and collecting contextual information. The proposed framework automates many of these routine tasks, allowing analysts to focus on strategic activities such as threat hunting and security planning.

7.3. Enhanced Scalability

As organizations continue to generate increasing volumes of security data, manually handling alerts becomes increasingly difficult. The distributed architecture of multiple AI agents enables the SOC to scale more effectively while maintaining operational performance.

7.4. Improved Decision Support

By combining threat intelligence, log analysis, and automated recommendations, the framework provides analysts with richer context for decision-making. Human analysts remain responsible for final security decisions, ensuring that critical actions remain under human oversight.

8. Future Implications

Future research can focus on integrating Large Language Models (LLMs), reinforcement learning, and autonomous response mechanisms into SOC operations. Additional studies can evaluate the framework in realworld enterprise environments and investigate secure agent-to-agent communication mechanisms.

Overall, the results demonstrate that a Multi-Agent AI architecture can significantly enhance SOC performance by reducing manual effort, accelerating incident response, and improving cybersecurity operations while maintaining human oversight and accountability.

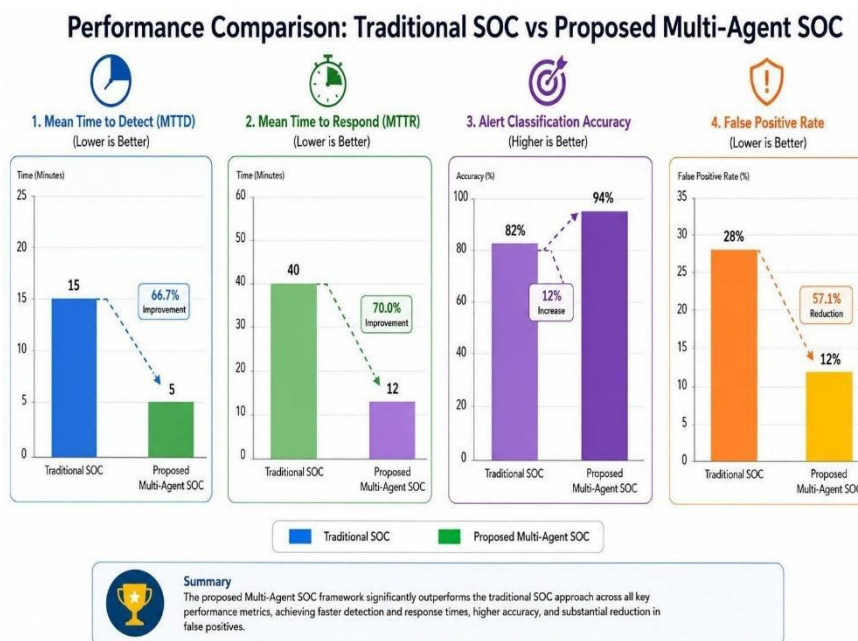


Figure 2.

9. Limitations and Future Work

A. Limitations

Although the proposed multi-Agent AI framework demonstrates significant potential for improving Security Operations Center (SOC) operations, several limitations must be considered.

1. Dependence on Data Quality

The effectiveness of AI agents largely depends on the quality, completeness, and accuracy of the input data. Incomplete logs, missing events, or outdated threat intelligence may lead to inaccurate analysis and recommendations.

2. False Positives and False Negatives

Despite advanced alert classification techniques, AI agents may incorrectly classify alerts. False positives can increase unnecessary investigations, while false negatives may allow genuine threats to remain undetected.

3. Limited Explainability

Many AI-based decision-making processes operate as black-box systems, making it difficult for analysts to fully understand how specific conclusions or recommendations were generated. This may reduce trust in automated decisions.

4. Dependence on External Threat Intelligence

The Threat Intelligence Agent relies on external threat intelligence sources for contextual information. Delays, inaccuracies, or unavailability of these sources can affect system performance.

B. Future Work

Several opportunities exist to enhance and extend the proposed Multi-Agent AI framework.

1. Real-World Deployment and Validation

Future research should implement and evaluate the framework in operational SOC environments to assess its effectiveness under real-world conditions and workloads.

2. Integration of Large Language Models (LLMs)

Advanced Large Language Models can be incorporated to improve threat analysis, incident summarization, natural language reporting, and decision support capabilities.

3. Autonomous Incident Response

Future versions of the framework may support controlled autonomous response actions such as endpoint isolation, malicious IP blocking, and automated remediation while maintaining appropriate human oversight.

4. Reinforcement Learning for Continuous Improvement

Reinforcement learning techniques can enable agents to learn from previous incidents and continuously improve.

10. Conclusion

Security Operations Centers (SOCs) play a vital role in protecting organizations against increasingly sophisticated cyber threats. However, traditional SOCs face significant challenges, including alert overload, analyst fatigue, slow incident response, fragmented security tools, and a shortage of skilled cybersecurity professionals. These limitations reduce operational efficiency and make it difficult for security teams to effectively manage the growing volume of security events.

This paper proposed a Multi-Agent Artificial Intelligence framework designed to enhance SOC operations through intelligent automation and collaborative decision-making. The proposed architecture consists of specialized agents, including an Alert Triage Agent, Threat Intelligence Agent, Investigation Agent, Response Agent, Reporting Agent, and a Coordinator Agent that orchestrates communication and workflow among agents. By distributing responsibilities across multiple autonomous agents, the framework enables efficient alert processing, threat enrichment, incident investigation, response recommendation, and automated reporting.

A simulation-based evaluation demonstrated the potential benefits of the proposed approach. The results indicate improvements in alert classification accuracy, reduction in false positives, faster incident detection and response times, and decreased analyst workload compared to traditional SOC workflows. These findings suggest that Agentic AI can significantly enhance the effectiveness, scalability, and responsiveness of modern cybersecurity operations.

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